Responsible Tech Innovation Through Design: A Participative, Reflective, and Systemic Approach

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Abstract
Technological progress has offered humanity many opportunities, though innovators need to gain knowledge in anticipating the impacts of their creations and dedicate themselves to more caring strategies, as many times technological advancements cause harm to the environment and people. In addition, organizations tend to respond to business externalities reactively, which has proven unsustainable. Understanding and identifying biases and expanding learning during the technological innovation process are essential to prevent unintended consequences. The future of responsible technology relies on achieving a more holistic view and enabling more systemic, sustainable, and responsible strategies. This study presents the formation of a substantive theory that explores the phenomena and development needs for designers successfully support responsibility in the technological innovation process. It reviews and compares designerly strategy models, methods, and frameworks. More concretely, it presents the process for defining a theoretical and methodological approach for a responsible, participatory, reflective, and systemic design.

Keywords
Responsible innovation
Co-creation
Innovation design
Sustainability design
Participatory design

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Introduction

Technology innovation has supported humanity in reaching Mars. However, when tech innovators intend to create solutions for society's future (Bethune, 2022), they must still take responsibility and commit to measuring technology-enabled solutions' impact on people of all colors and cultures and the planet's diversity and harmony.

There have been adverse effects of technological advancement, many well-recognized and others still topic sensitive and challenging to debate (Costanza-Chock, 2020). For example, Technology has been criticized for perpetuating societal injustice and harmful biases (Friedman et al., 2017). Colonial legacies about the dominating North's political, material, and technological superiority (Parashar & Schulz, 2021) have also incepted prejudice and hate that remains.

On the other hand, human curiosity and willingness to build a better, just, and fair society are maturing our understanding of human rights, regulations, and environmental and biodiversity comprehension. In 2021 and 2022, the primary author collected extensive literature on Responsible and Sustainable Innovation and Design from social media.

The content reveals increased demand for more responsible and sustainable solutions and outcomes. Design, Business, and Technology, academics, and practitioners share, research, and seek processes, practices, methods, principles, and examples of suitable applications and tools that can fulfill these needs.

By using a grounded theory approach to this contemporary sample, this study aims to present the initial formation of a substantive theory on exploring the phenomena of Design practice in creating and developing sustainable and responsible technological outcomes. It proposes practical development needs that define principles for designers to successfully support innovators to take responsibility for technological solutions and operations. More concretely, this paper presents the process of defining a theoretical and methodological approach for a responsible, participatory, reflective, and systemic design.

Research Approach

The study relies on a grounded theory approach (Charmaz, 2014) consisting of various iterations until the desired outcome is achieved. As a preliminary phase of the theory construction, in 2021 and 2022, a documents analysis was made of a total of 95 professional and academic articles and case studies shared on LinkedIn, Twitter, and by Business and Design communities on Slack, such as Service Design Network, Design Justice, IxDA, All Tech is Human, Design Activism Group and Speculative Futures social media groups and channels. Of this sample, ten design methods and design strategies were selected for more detailed analysis. The selection was based on these methods and strategies' general popularity and familiarity. Furthermore, they included ethicality as an emerging topic. In the thematic analysis, during the coding process, we identified themes that permitted the critical points of the data to be gathered considering Humans, Nature, Technology, Governance & Business, Collabora-

tion, and Design Scope. Once the initial coding was finalized, purposive sampling was conducted on a Miro board, including selected topic-related academic articles to understand these themes further. Memoing supported the constant comparative analysis and inductive analytical process of the data collection while contrasting data, concepts, categories, and theoretical claims while keeping constructive knowledge (Charmaz, 2014).

The generation of a substantive has a broader research objective "Explore the phenomena of designers involving and engaging people in sustainable and responsible technological innovations through Design practice". Constructivism-grounded theory (CGT) was primarily selected to construct a substantive theory using reflexive analysis and abductive reasoning (Charmaz, 2014), in which researchers' personal beliefs align with constructivist assumptions. We are currently working with the theory construction based on theoretical sampling, desktop research, and data coding. Next, data collection will be extended to interviews and focus groups with experts in Design. After that, the theory will be co-designed and validated with diverse members of design and business communities. Finally, sustainable and responsible innovation facilitation and design tools will be co-created, selected, tailored, validated, and launched.

Results

The following chapters first present an overview of the sample collection and comparison criteria. Second, the authors interpret the changes in the Design discipline in terms of maturity and scope. Third and fourth, it proposes enhancements in the technological innovation process.

Formulating substantive theory for designing technological innovation

Most recent studies in Design highlight the importance of addressing technological innovation's environmental and societal implications (Gaziulusoy & Öztekin, 2019). Considering these implications demands expertise in areas that are only sometimes familiar to technology experts or companies that follow a tech-driven innovation process. Moreover, the lack of knowledge about societal and environmental aspects linked to technologies can lead to unintended and unpredicted consequences.

Design can play a role in fostering transdisciplinary (Coulter, 2013) approaches to tackle these complex challenges. Teams of diverse experts engage in a co-creative and participatory process, benefiting innovation facilitation, methods, and tools that support people's engagement and sense of responsibility towards distributed decisions and commitments to achieve short- and long-term goals with actants in the system.

Design theories and methodologies diverge and follow similar criteria. We used these emerging themes from the theory formation to highlight in this research for comparison, followed by supportive questions:

- Human.
  How is human centricity applied? How do identity intersections and pluralism are applied? How values and ethical boundaries are considered? What is the human relationship with other systems?
- Nature.
  How is the natural environment considered? What is the proposed vision for natural recovery? How does it suggest mitigating current and future environmental challenges?
- Technology.
  How does it support technological requirements and implementation? What is the role of technology in the innovation process? How does it relate to the other systems?
- Governance.
  How does it support solution creation and execution? How does it support mindset change and cultural and operational organization transformation? Does it support the decision-making process? Does it build certainty?
- Collaboration.
  What are the bases for collaboration? How does it support engagement, accountability, commitment, transdisciplinary, and participation?
- Design Dimension. What is the Design scope? What kind of outcomes are the approaches targeting?

Expanding Design Scope and vision maturity

Growing digitalization is causing an exponential increase in designed products and services. That is why Design is moving towards more intangible and strategic application contexts, and its scope and key aims have broadened (Buchanan, 2007). The Design discipline has also evolved reactively rather than proactively.

For instance, in Design Thinking (Kimbell, 2011), Design is a reflexive act that helps to understand wicked problems in ambiguous spaces. It assumes that the outcomes have multiple meanings depending on the individuals. Tab. I shows the results of categorizing dimensions of Design Thinking in six emerging themes in the literature based on authors such as Brown (2009), Kelley and Kelley (2013), and Kimbell (2014).

Human	Nature	Technology	Governance	Collaboration	Design dimension
Searches for tacit and explicit knowledge from prospects and current users of a design process outcome—By empathy, generate more human-centric solutions.	It needs to be intentionally inserted into the thinking. Most authors understand the design's impact on individuals and the environment.	Early interactions with the solutions (i.e., sketches and prototypes) can result in better models (Henry Petroski). It approaches technologies with their characteristics and constraints to modelling content and, often, knowledge manipulations. (How something should be understood)	Ensures the success of the outcomes by involving customers in the design process —customer empathy, customer-centric.	It values human engagement as the ethics of the design process. The understanding of complex problems of design can be improved by demanding participation.	*Design* is a reflexive act that helps understand wicked problems (exists in an ambiguous space). It assumes that the outcomes of the Design have multiple meanings to multiple individuals.

Tab. I
Dimensions of Design Thinking categorized based on comparison themes and supporting questions.

Service Design proposes integrating organizational un-siloed thinking towards customer touchpoints. Touchpoints are moments of truth and crucial interactions that support customer-organization interactions. Value is created and delivered throughout the service ecosystem and in a continuous and direct dialogue. Tab. II presents the results of categorizing dimensions of Service Design in six emerging themes (e.g., Sanders & Stappers, 2008; Moritz, 2005; Ramaswamy, 2009; Stickdorn et al., 2018).

Human	Nature	Technology	Governance	Collaboration	Design dimension
In addition to the original ideas from Design Thinking which focus on customer/user experience, SD considers services as temporal economic activities that happen over time from the experience life cycle.	Service Design purpose supports ecosystems strategies. Companies in a value network and value chain work together to achieve customer excellence. If targets are ecological targets. Service Design can help to achieve them.	Digital transformation supports creating many digital services and systems to support customer-centric operations. Digital channels are limited and controlled ambients where designers can create consistency and affordances to ease the use and automate processes.	Service Design is a bridge for multidisciplinary work and operations alignment toward customer-lifecycle experiences.	Co-creation is at the core of Service Design, where service and product improvements and innovation are based on the customer's understanding and demand and multidisciplinary teams operating together to fulfil these needs.	Service Design has a strategic approach and tends to take a bird's eye to the service experience life cycle, value chain, and organisational operations; It can also be considered practical as the nature of the work is to find gaps and areas for improvements prioritised towards

Tab. II
Dimensions of Service Design categorized based on comparison themes and supporting questions.

Circular Design uses the Design Thinking approach to explore new ways to create sustainable, resilient, long-lasting value in the circular economy – giving creative confidence to redesign products, processes, operations, strategies, businesses, or even cities. Dimensions of Circular Design were grouped under six emerging themes in the literature (e.g., Takacs et al., 2020; Meloni et al., 2018; MacArthur, 2021), as presented in Tab. III.

Human	Nature	Technology	Governance	Collaboration	Design dimension
Adopting sustainable and responsible business models and deploying effective reverse cycles is instrumental to realising greater product circularity. Consumers continue to landfill their products after the first use or store them in a closet forever; the impact of better Design is limited. Therefore, there is a need to drive customer behaviour to build an emotional attachment for their devices to have a longer life, and when changing to a new one, the device does not go on the wrong path.	Nature has a passive role and is seen as a victim of our progressive actions and extensive exploitation of resources, pollutive processes, and industry. Climate change, natural disasters, and scarcity of resources call for businesses to act. Some authors refer to the need to regenerate natural systems to sequester carbon in soil and products. They also say that a Circular Economy is a framework that tackles global challenges such as climate change, biodiversity loss, waste and pollution.	Devices are part of our life, but most are still disposable as new models offer more capabilities than the old ones. These disposal practices lead to the loss of energy, resources, and product value embodied in electronics, generating a vast amount of waste in the process. New devices should contain used and remanufactured components and work as a gateway to cloud adaptability in computing power and memory allocation. Maintenance services to ensure longer life investments in the recycling process to ensure that materials are recovered and reused. Some authors argue that advances in AI, cloud and fog computing can accelerate the transition towards a circular economy at scale.	Rethinking the Design of products is essential to enabling the economic reuse of assets and their components and materials. Shifting businesses to circular operations and product design / inner loops Value and use of components can also be cascaded across industries, increasing investments in material renewal / outer loops. Staffing for the planet is necessary, and new roles and responsibilities tackle and orchestrate circular operations and outcomes. Increasing circular policies - working in conjunction - helps governments build healthier economic recoveries and lower the cost of transition for businesses.	Collaboration also happens in the supply chain/value network as solid partnerships, and ecosystem strategies enable circular practices. Circular Design requires product design, management, operations, and supply chain changes. There is an excellent push for value co-creation internally in the enterprise and at the business network levels.	Circular design strategies fit the business model and the broader system in which the device operates.

Tab. III
Dimensions of Circular Design categorized based on comparison themes and supporting questions.

Regenerative Design focuses mainly on the co-evolution of two systems, sociocultural and ecological (Cole, 2012). Derived from global and local, glocal recognizes the need to balance the invisible global forces and the real sense of place and culture (Nagashima, 1999). It also characterizes spaces as places with heritage, cultural and national meanings.

Furthermore, other Design approaches were analyzed using the emerging themes and supporting questions. They entailed Systemic Design, Design for Sustainability Transitions, Value Sensitive Design, Design Justice, Biomimicry, Design Council's Beyond Net-Zero, and business strategies such as Planet Boundaries, Doughnut Economics, Sustainable Development Goals, and Wedding Cake, Ten Types of Innovation, Responsible Innovation Framework and Resilient and Regenerative Cultures.

The studies reveal that awareness throughout the design process is growing, embracing a post-human-centric worldview. Still, technological innovation depends on a continuous decision process, often limited and structured by a biased group of wealthy individuals. It is also not enough to think about digital experiences or circular Design in technological innovation; we need to step further in the regeneration of systems to nourish a healthier planet and a more fair and just society. Fig. 1 visualizes our proposal for extending the Design scope and maturity from a human-centric view focusing on ergonomics or semantics, interactions, and experiences to a post-human-centric one considering harmonization and regeneration. Such a worldview acknowledges and builds upon the learnings of the past: ever-growing knowledge and competence enrich the role of Design. Its starting point was Buchanan's (1992) four orders of Design which refer to the different levels of design focus: symbols, products, services and systems, and the other pieces of material in this study. It also uses questions as an example of changing mindset and maturity of Design to tackle the real-life diversity and complexity and impact of innovation on the planet and society.

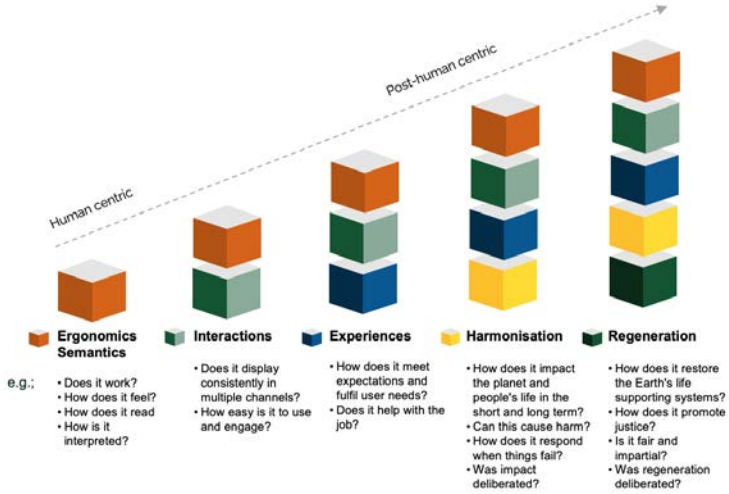


Fig. 1
Scaling systems awareness across the design scope. Credits: Jane Vita.

Technology as an Enabler for Transformation

Technology has the potential to enable humans to mitigate global challenges. However, if it is not ethically and responsibly implemented, there may be unexpected and unintended consequences, as it follows a paradigm that perpetuates societal injustice and harmful biases. This paradigm must shift from mono-disciplinary and homogenous knowledge production to more collaborative, diverse, and participatory practices (Liao & Muller, 2019).

Regardless of the type of design intervention, be it a product or a service, it is fundamental to view it in its larger context, i.e., the systems it is entangled. The systemic view will pose new questions on who or what is in relationship with the Design; therefore, who should be considered an actant? The systems view crossover design interventions, from stuff to strategy, and technological innovation does not work in isolation. It can only be implemented by considering its connections and impact on other realms, such as society and the biosphere.

The attributes of technological innovation must go beyond human-centricity. Fig. 2 shows our expanded User Experience Hierarchy of Needs model based on Anderson (2011). His model is highly used in human-computer interaction design to define the standard maturity experience we want people to have. The expansion proposed based on the purpose sampling encompasses inclusion, harmonic, and regenerative maturity levels where the charm now is not cross-convenience level but goes beyond human centricity.

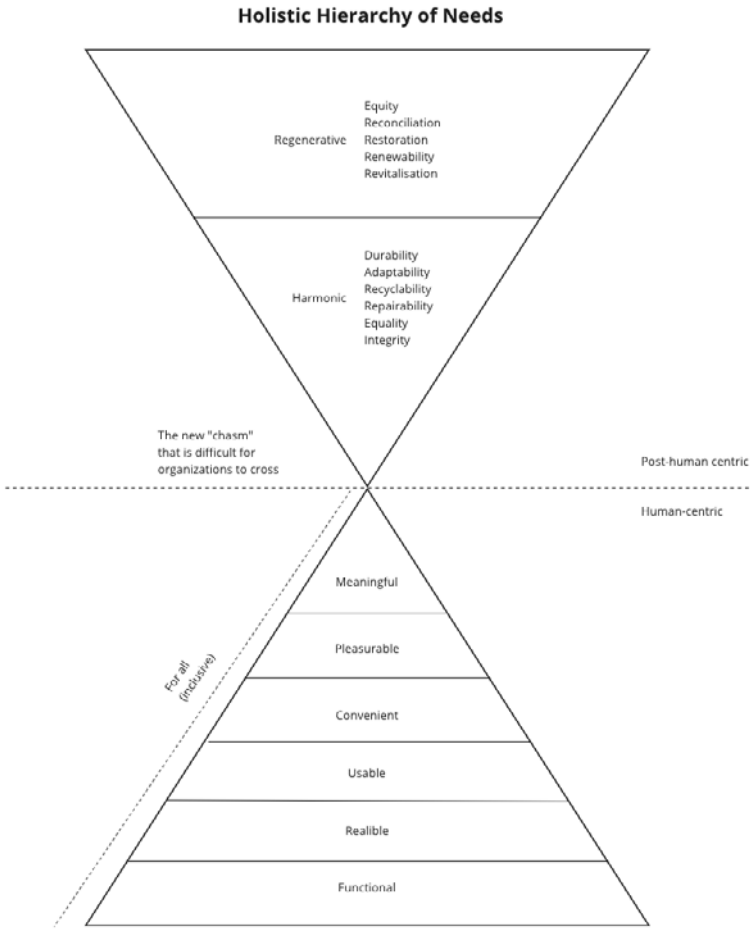


Fig. 2
Expansion of the *Experience Hierarchy of Needs* (Anderson, 2011) to a more holistic needs view. Credits: Jane Vita.

The bottom part of the pyramid focuses on tasks, functions of the product or service, and features that make it work. Includes tasks that are easy to measure and quantify with straightforward objectives. Towards the top of the bottom pyramid, middle of the diagram, the focus shifts to experiences, which are more challenging to measure as they have more personal significance, requiring a better understanding of people, activities, and context. In the original diagram and explanation presented by Anderson, different people experience different things. However, it does not mention diversity in designing these technological products and services to be more inclusive. We propose special attention to inclusion in this reviewed diagram, from function to meaningful level.

Adding an upside-down pyramid to the diagram shows that designing technological solutions should go beyond human needs and reach needs at the systems levels. Starting from ensuring Harmony, not causing harm, and respecting the existing systems in play to a more just and fair Regenerative impact, reconciling and reviving when opportunities occur. In addition to the levels presented in Fig. 2, human rights and topics such as data access, privacy, and transparency should be considered.

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Thinking and acting on needs beyond human-centrism will help organizations uncover needs that will support a positive impact and avoid harm to connected systems, such as the biosphere and society. It is not only about how a product or service functions, is reliable, useful, and experienced, but also about how its production impacts, e.g., employees, customers, and society in the long term, and how it supports health and privacy. It is about how a culture of care and responsibility is embedded in the technological solution process and ways of working.

Dedicating more time to understanding and acting toward a more responsible process and outcomes is needed, as well as measuring and monitoring the ongoing services and product updates and their continued use. As in the ancient Haudenosaunee (Iroquois), The Seventh Generation Principle philosophy (Redvers et al., 2020) declares, "Decisions we make today should result in a sustainable world seven generations into the future."

At the core of technological innovation are *Agile* and *Lean* methodologies, and their rapid approach does not allow spending enough time considering things beyond the necessary to build. They advocate for a culture of failure but are limited to a biased group that might not intentionally advocate for diverse perspectives and a participatory approach.

Expansive Learning through the sustainable and responsible innovation journey

Responsible businesses foster a caring culture in operations, which leads to more sustainable outcomes. It nurtures interdisciplinary and participatory collaboration with actants in systems impacted by their solutions (Jones, 2014). It acknowledges that people leading businesses and changing mindsets are crucial to contributing to a healthier planet and society.

People involved in the innovation process must be committed, resourceful and skillful, and facilitators ensure this knowledge is diversely produced and shared (Drain & Sanders, 2019), communicated, and embodied in technology design. By cooperating and forming boundary objects (Välk et al., 2020), we provide spaces for sharing learning and skills and inspiring future states.

Methods and tools should support systemic thinking with approaches and clues that consider aspects of diversity, inclusion, planet boundaries, and impact, which people would not access otherwise. The initial phases are recognizing and discovering the systems and actants involved, their relationships, value exchange, and possible implications. Creating constraints and criteria for technological concept creation, solution development, and business strategies with the foundation of sustainable and responsible values is essential to achieving these outcomes.

Expanding and producing knowledge through Design should happen in an iterative, reflective, participative, and respectful process. Fig. 3 exemplifies a more holistic and purposeful Innovation Journey, where the path grows toward care and certainty. Broadening interests in affected systems is necessary to achieve sustainable and responsible results.

In this approach, the reflective and participatory innovation journey begins with planning, mapping, and understanding, identifying focus areas, and reframing the problem. Context is explored through different lenses, considering technological, economic, cultural, societal, and environmental views. In the analysis and framing phase, key topics from each lens are translated into criteria. In describing phase, priority criteria are selected, and the value proposition is reframed. Next, lead to ideating approaches and technological solutions. Turning criteria become a source for creation, and rapid prototyping and validation can begin. In the refining and impact assessment phase, solution requirements are defined. Finally, requirements are tested in user experience testing, and expert evaluations lead to scoping and prioritizing requirements and defining key performance indicators (KPIs).

To complement this, preliminary work on designing the process is crucial (Vines et al., 2013). Participation becomes configurable as an interventionist and reflective practice where new ideas, processes, and perspective lenses are introduced to encourage mindset change and promote dialogue. Configuring participation and designing the engagement throughout the technological innovation process can support expansive learning (Engeström, 2001) and dialogue around responsible and sustainable practices and outcomes. It is essential to create contact zones (Pratt, 1991) and involve affected communities to future-proof outcomes, where technology is an enabler, not the driver.

We understand that an expansive transformation through learning is required to reach collective responsibility toward new patterns and breaking existing paradigms, which will purposely impact the innovation journey and its outcomes.

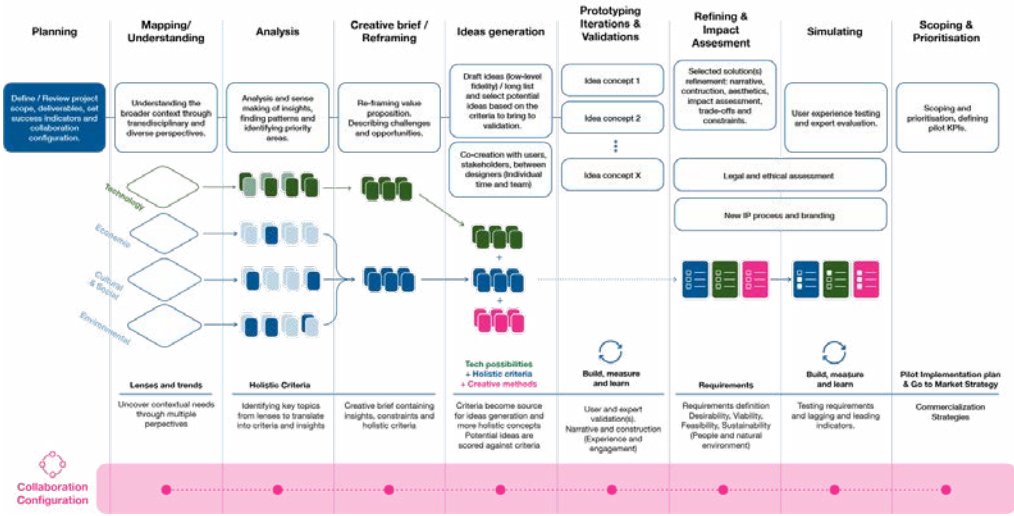


Fig. 3
Reflective and participatory innovation process. Credits: Jane Vita.

Conclusions

This paper described the formulation of substantive theory and approach for a responsible, participatory, reflective, and systemic design. Further studies will enhance the substantive theory with validation interviews and workshops, including diverse actants, as mentioned previously in the configuration participation of the design research framework. We also intend to validate tailored creative material that can serve as a toolkit to expand knowledge and diverse perspectives in the responsible innovation process. As an outcome, we expect a practical and tailored approach and tools to support the role of Design in technological innovation. The aim is to continuously investigate different phases of the innovation journey to inform, reflect, and produce diverse knowledge to create more responsible and sustainable solutions with humans and nature at the core, supporting impact visualization, risk assessment, and mitigation.

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